

# Git Assignment

## Assignment :2(a)

Create version control account on GitHub and using Git Commands to Create a repository and push your code on github.

### What is Git?

□ Git is a free, open-source version control system (VCS) that helps you track changes to your code and projects

□ It was created by Linus Torvalds in 2005, and has been maintained by Junio Hamano since then.

□ Features:

1. Creates backups

2. Supports non-linear development

3. Supports collaboration

### What is GitHub?

□ GitHub is a website that lets developers store, manage, and share their code.

□ GitHub was founded in 2008 and has been a subsidiary of Microsoft since 2018.

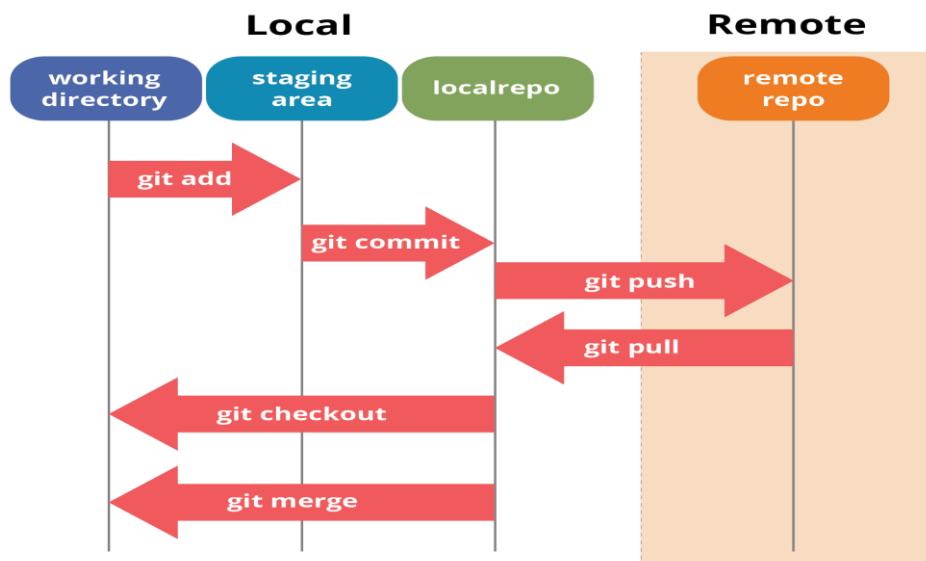
Microsoft acquired GitHub for \$7.5 billion

□ GitHub also offers tools for collaboration, such as bug tracking, task management.

### Difference of Git And GitHub :

| Git                                           | GitHub                                                        |
|-----------------------------------------------|---------------------------------------------------------------|
| It is a version control system.               | It is a cloud platform for hosting Git repositories.          |
| Works locally to track code changes.          | Stores and manages Git repositories online.                   |
| Does not require the internet to work.        | Requires the internet to access repositories.                 |
| Helps in managing different versions of code. | Allows collaboration, issue tracking, and project management. |
| Open-source and free to use.                  | Owned by Microsoft, offers free and paid plans.               |

## Git Architecture:



## Component of Git :

- ❑ **Working Directory:** This is where your project files are stored and modified before tracking them with Git. Changes here are untracked until added to the staging area.
- ❑ **Staging Area:** A temporary storage where files are added using `git add` before committing. It helps organize changes before finalizing them in the repository.
- ❑ **Local Repository:** A Git repository stored on your computer where committed changes are saved. You can work offline and push updates to a remote repository later.
- ❑ **Remote Repository:** A shared repository (like on GitHub) that allows multiple developers to collaborate by pushing and pulling changes.

## Git Commands:

### Before Push:

The screenshot shows the GitHub repository setup page. It includes sections for setting up GitHub Copilot, adding collaborators, and quick setup instructions. The quick setup section provides a command line sequence for creating a new repository.

```
echo "# WAD-" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/Asmita362/WAD-.git
git push -u origin main
```

```

MINGW64/c/Users/acer/Desktop/WAD_LAB/Assignment_3
acer@DESKTOP-KV09KVS MINGW64 ~
$ cd C:/Users/acer/Desktop/WAD_LAB/Assignment_3
acer@DESKTOP-KV09KVS MINGW64 ~/Desktop/WAD_LAB/Assignment_3
$ git init
Initialized empty Git repository in C:/Users/acer/Desktop/WAD_LAB/Assignment_3/.git/
acer@DESKTOP-KV09KVS MINGW64 ~/Desktop/WAD_LAB/Assignment_3 (master)
$ git add
warning: in the working copy of 'form1.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'form1.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'users.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'users.js', LF will be replaced by CRLF the next time Git touches it
acer@DESKTOP-KV09KVS MINGW64 ~/Desktop/WAD_LAB/Assignment_3 (master)
$ git commit -m "Assignment_3"
[master (root-commit) b67dcd] Assignment_3
4 files changed, 82 insertions(+)
create mode 100644 form1.html
create mode 100644 form1.js
create mode 100644 users.html
create mode 100644 users.js
acer@DESKTOP-KV09KVS MINGW64 ~/Desktop/WAD_LAB/Assignment_3 (master)
$ git branch -M main
acer@DESKTOP-KV09KVS MINGW64 ~/Desktop/WAD_LAB/Assignment_3 (main)
$ git remote add origin https://github.com/LakshanaBachhav25/WAD_LAB.git
acer@DESKTOP-KV09KVS MINGW64 ~/Desktop/WAD_LAB/Assignment_3 (main)
$ git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.72 KiB | 351.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/LakshanaBachhav25/WAD_LAB.git
 * (new branch) main -> main
branch 'main' set up to track 'origin/main'.
acer@DESKTOP-KV09KVS MINGW64 ~/Desktop/WAD_LAB/Assignment_3 (main)
$ |

```

## After Push:

**WAD-Practical**
Public
Pin
Unwatch
1

main
1 Branch
0 Tags

Add file
Code

|  |                                                               |                                      |              |
|--|---------------------------------------------------------------|--------------------------------------|--------------|
|  | <b>VaishnaviPatole02</b> Moved public folder into Assignment4 | 95c893b · 2 weeks ago                | 9 Commits    |
|  | .vscode                                                       | initial commit                       | last month   |
|  | Assignment1                                                   | Add files via upload                 | 2 months ago |
|  | Assignment3                                                   | Add files via upload                 | 2 months ago |
|  | Assignment4                                                   | Moved public folder into Assignment4 | 2 weeks ago  |
|  | .editorconfig                                                 | initial commit                       | last month   |
|  | .gitignore                                                    | initial commit                       | last month   |
|  | README.md                                                     | initial commit                       | last month   |

## Before Pull:

WAD-Practical / Assignment3 /
Add file
...

**VaishnaviPatole02** Add files via upload
 2001497 · 2 months ago
 History

| Name          | Last commit message  | Last commit date |
|---------------|----------------------|------------------|
| ..            |                      |                  |
| ajax1.html    | Add files via upload | 2 months ago     |
| ajax2.html    | Add files via upload | 2 months ago     |
| register.html | Add files via upload | 2 months ago     |
| register.js   | Add files via upload | 2 months ago     |
| userlist.html | Add files via upload | 2 months ago     |
| userlist.js   | Add files via upload | 2 months ago     |

```

acer@DESKTOP-KV09KVS MINGW64 ~/Desktop/WAD_LAB/Assignment_3 (main)
$ git fetch origin

acer@DESKTOP-KV09KVS MINGW64 ~/Desktop/WAD_LAB/Assignment_3 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

acer@DESKTOP-KV09KVS MINGW64 ~/Desktop/WAD_LAB/Assignment_3 (main)
$ git log origin/main
commit b67dcccdb128c1ceae814f16644851cad903299f (HEAD -> main, origin/main)
Author: LakshanaBachhav25 <lakshanabachhav25@gmail.com>
Date:   Sun Feb 16 20:59:38 2025 +0530

    Assignment_3

acer@DESKTOP-KV09KVS MINGW64 ~/Desktop/WAD_LAB/Assignment_3 (main)
$ git pull origin
Already up to date.

```

## After Pull:

| WAD-Practical / Assignment3 /  |                      |                  | Add file ▾             | ...                                                                                         |
|-----------------------------------------------------------------------------------------------------------------|----------------------|------------------|------------------------|---------------------------------------------------------------------------------------------|
|  VaishnaviPatole02             | Add files via upload |                  | 2001497 · 2 months ago |  History |
| Name                                                                                                            | Last commit message  | Last commit date |                        |                                                                                             |
|  ..                            |                      |                  |                        |                                                                                             |
|  ajax1.html                    | Add files via upload | 2 months ago     |                        |                                                                                             |
|  ajax2.html                    | Add files via upload | 2 months ago     |                        |                                                                                             |
|  register.html                 | Add files via upload | 2 months ago     |                        |                                                                                             |
|  register.js                   | Add files via upload | 2 months ago     |                        |                                                                                             |
|  userList.html                | Add files via upload | 2 months ago     |                        |                                                                                             |
|  userList.js                 | Add files via upload | 2 months ago     |                        |                                                                                             |